of the students using SAT preparatory materials from the three major publishers of those materials. Thus, we need to develop a statistical test that will allow us to compare $t > 2$ population variances. We will consider two procedures. The first procedure, Hartley's test, is very simple to apply but has the restriction that the population distributions must be normally distributed and the sample sizes equal. The second procedure, Levine's test, is more complex in its computations but does not restrict the population distributions or the sample sizes. Levine's test can be obtained from many of the statistical software packages. For example, SAS and Minitab both use Levine's test for comparing population variances.

Hartley F_{max} test

H. O. Hartley (1950) developed the **Hartley F_{max} test** for evaluating the hypotheses

$$H_0\colon \sigma_1^2 = \sigma_2^2 = \ldots = \sigma_t^2 \text{ vs. } H_a\colon \sigma_i^2\text{s not all equal}$$

The Hartley F_{max} requires that we have independent random samples of the same size n from t normally distributed populations. With the exception that we require $n_1 = n_2 = \ldots = n_t = n$, the Hartley test is a logical extension of the F test from the previous section for testing $t = 2$ variances. With s_i^2 denoting the sample variance computed from the ith sample, let s_{min}^2 = the smallest of the s_i^2s and s_{max}^2 = the largest of the s_i^2s. The Hartley F_{max} test statistic is

$$F_{max} = \frac{s_{max}^2}{s_{min}^2}$$

The test procedure is summarized here.

Hartley's F_{max} Test for Homogeneity of Population Variances

H_0: $\sigma_1^2 = \sigma_2^2 = \ldots = \sigma_t^2$ homogeneity of variances

H_a: Population variances are not all equal

T.S.: $F_{max} = \dfrac{s_{max}^2}{s_{min}^2}$

R.R.: For a specified value of α, reject H_0 if F_{max} exceeds the tabulated F value (Table 12) for $a = \alpha$, t, and $df_2 = n - 1$, where n is the common sample size for the t random samples.

Check assumptions and draw conclusions.

We will illustrate the application of the Hartley test with the following example.

EXAMPLE 7.8

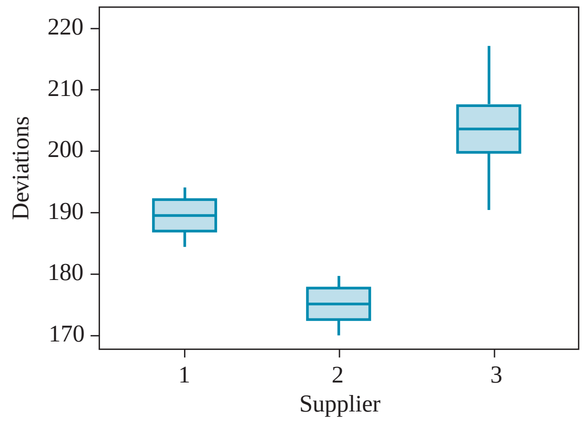
Wludyka and Nelson [*Technometrics* (1997), 39: 274–285] describe the following experiment. In the manufacture of soft contact lenses, a monomer is injected into a plastic frame, the monomer is subjected to ultraviolet light and heated (the time, temperature, and light intensity are varied), the frame is removed, and the lens is hydrated. It is thought that temperature can be manipulated to target the power (the strength of the lens), so interest is in comparing the variability in power. The data are coded deviations from target power using monomers from three different suppliers. We wish to test $H_0\colon \sigma_1^2 = \sigma_2^2 = \sigma_3^2$.

Deviations from Target Power for Three Suppliers

	Sample										
Supplier	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	n	s_i^2
1	191.9	189.1	190.9	183.8	185.5	190.9	192.8	188.4	189.0	9	8.69
2	178.2	174.1	170.3	171.6	171.7	174.7	176.0	176.6	172.8	9	6.89
3	218.6	208.4	187.1	199.5	202.0	211.1	197.6	204.4	206.8	9	80.22

Solution Before conducting the Hartley test, we must check the normality condition. The data are evaluated for normality using a boxplot given in Figure 7.12.

FIGURE 7.12
Boxplot of deviations from target power for three suppliers



All three data sets appear to be from normally distributed populations. Thus, we will apply the Hartley F_{max} test to the data sets. From Table 12, with $a = .05$, $t = 3$, and $df_2 = 9 - 1 = 8$, we have $F_{max,.05} = 6.00$. Thus, our rejection region will be

R.R.: Reject H_0 if $F_{max} \geq F_{max,.05} = 6.00$

$$s^2_{min} = \min(8.69, 6.89, 80.22) = 6.89$$

and

$$s^2_{max} = \max(8.69, 6.89, 80.22) = 80.22$$

Thus,

$$F_{max} = \frac{s^2_{max}}{s^2_{min}} = \frac{80.22}{6.89} = 11.64 > 6.00$$

Thus, we reject H_0 and conclude that the variances are not all equal.

If the sample sizes are not all equal, we can take $n = n_{max}$, where n_{max} is the largest sample size. F_{max} no longer has an exact level α. In fact, the test is liberal in the sense that the probability of Type I error is slightly more than the nominal value α. Thus, the test is more likely to falsely reject H_0 than the test having all n_is equal when sampling from normal populations with the variances all equal.

The Hartley F_{max} test is quite sensitive to departures from normality. Thus, if the population distributions we are sampling from have a somewhat nonnormal distribution but the variances are equal, the F_{max} will reject H_0 and declare the variances to be unequal. The test is detecting the nonnormality of the population distributions, not the unequal variances. Thus, when the population distributions are nonnormal, the F_{max} is not recommended as a test of homogeneity of variances. An alternative approach that does not require the populations to have normal distributions is the Levine test. However, the Levine test involves considerably more calculation than the Hartley test. Also, when the populations have a normal distribution, the Hartley test is more powerful than the Levine test. Conover, Johnson, and Johnson [*Technometrics,* (1981), 23: 351–361], conducted a simulation study of a variety of tests of homogeneity of variance, including the Hartley and Levine test. They demonstrated the inflated α levels of the Hartley test when the populations have highly skewed distributions and recommended the Levine test as one of several alternative procedures.

The Levine test involves replacing the jth observation from sample i, y_{ij}, with the random variable $z_{ij} = |y_{ij} - \tilde{y}|$, where $\tilde{y}$ is the sample median of the ith sample. We then compute the Levine test statistic on the z_{ij}s.

Levine's Test for Homogeneity of Population Variances

H_0: $\sigma_1^2 = \sigma_2^2 = \ldots = \sigma_t^2$ homogeneity of variances

H_a: Population variances are not all equal

T.S.: $$L = \frac{\sum_{i=1}^{t} n_i(\bar{z}_{i.} - \bar{z}_{..})^2/(t-1)}{\sum_{i=1}^{t}\sum_{j=1}^{n_i}(z_{ij} - \bar{z}_{i.})^2/(N-t)}$$

R.R.: For a specified value of α, reject H_0 if $L \geq F_{\alpha,\text{df}_1,\text{df}_2}$, where $\text{df}_1 = t - 1$, $\text{df}_2 = N - t$, $N = \sum_{i=1}^{t} n_i$, and $F_{\alpha,\text{df}_1,\text{df}_2}$ is the upper α percentile from the F distribution, Table 8.

Check assumptions and draw conclusions.

We will illustrate the computations for the Levine test in the following example. However, in most cases, we would recommend using a computer software package such as SAS or Minitab for conducting the test.

EXAMPLE 7.9

Three different additives that are marketed for increasing the miles per gallon (mpg) for automobiles were evaluated by a consumer testing agency. Past studies have shown an average increase of 8% in mpg for economy automobiles after using the product for 250 miles. The testing agency wants to evaluate the variability in the increase in mileage over a variety of brands of cars within the economy class. The agency randomly selected 30 economy cars of similar age, number of miles on their odometer, and overall condition of the power train to be used in the study. It then randomly assigned 10 cars to each additive. The percentage increase in mpg obtained by each car was recorded for a 250-mile test drive. The testing agency wanted to evaluate whether there was a difference between the three additives with respect to their variability in the increase in mpg. The data are give here along with the intermediate calculations needed to compute the Levine's test statistic.

Solution Using the plots in Figures 7.13 (a)–(d), we can observe that the samples from additive 1 and additive 2 do not appear to be samples from normally distrib-